Project Report: Custom UNIX Utilities

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Course: Operating Systems Lab (CSC211)

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1. Project Overview

This project involves the implementation of seven core UNIX command-line utilities from scratch using the **C11** standard. The goal was to understand low-level system calls, file I/O, and directory handling in a POSIX-compliant environment. To prevent conflicts with existing system binaries, all utilities are prefixed with `custom_`.

Utilities Implemented:

- `custom_ls`: Directory listing with support for permissions, hidden files, and sorting.
- `custom_cat`: File concatenation and display with advanced formatting flags.
- `custom_grep`: Pattern matching using string manipulation and recursive directory traversal.
- `custom_wc`: Counting lines, words, and bytes.
- `custom_cp`: File and directory duplication.
- `custom_mv`: File renaming and moving.
- `custom_rm`: Secure file and directory deletion.

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2. Technical Specifications

- **Language:** C (C11 Standard)
- **Compiler:** GCC
- **Build System:** GNU Make
- **Libraries used:** `stdio.h`, `stdlib.h`, `unistd.h`, `dirent.h`, `sys/stat.h`, `string.h`

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3. Implementation Details & Screenshots

3.1 custom_ls

Functionality: Lists files in a directory. Implements `-l` (long format), `-a` (all files), and `-h` (human-readable).

```
debashishbhagawati@Debashishs-MacBook-Pro utils_test % custom_ls -lah
total 8
drwxr-xr-x  6 debashishbhagawati staff         192 Apr 12 16:33 .
drwx------ 38 debashishbhagawati staff        1.2K Apr 12 16:33 ..
-rw-r--r--  1 debashishbhagawati staff           0 Apr 12 16:33 .hidden_config
-rw-r--r--  1 debashishbhagawati staff          97 Apr 12 16:33 fileA.txt
-rw-r--r--  1 debashishbhagawati staff          30 Apr 12 16:33 fileB.txt
drwxr-xr-x  3 debashishbhagawati staff          96 Apr 12 16:33 nested_dir
debashishbhagawati@Debashishs-MacBook-Pro utils_test % custom_ls -lrt
total 8
drwxr-xr-x  3 debashishbhagawati staff          96 Apr 12 16:33 nested_dir
-rw-r--r--  1 debashishbhagawati staff          30 Apr 12 16:33 fileB.txt
-rw-r--r--  1 debashishbhagawati staff          97 Apr 12 16:33 fileA.txt
```

3.2 custom_cat

Functionality: Reads files and outputs to stdout. Supports `-n` (line numbering) and `-A` (visualizing tabs and line ends).

```
debashishbhagawati@Debashishs-MacBook-Pro utils_test % custom_cat -nsA fileA.txt
     1  Line 1: Hello World$
     2  Line 2: UNIX is awesome$
     3  $
     4  Line 4:^IThis line has a tab.$
     5  Line 5:   Spaces here.$
debashishbhagawati@Debashishs-MacBook-Pro utils_test % custom_cat fileB.txt fileA.txt
Just a simple secondary file.
Line 1: Hello World
Line 2: UNIX is awesome

Line 4: This line has a tab.
Line 5:   Spaces here.
```

3.3 custom_grep

Functionality: Searches for a specific string pattern. Supports `-i` (case-insensitive) and `-r` (recursive).

```
debashishbhagawati@Debashishs-MacBook-Pro utils_test % custom_grep -in "unix" fileA.txt
2:Line 2: UNIX is awesome
debashishbhagawati@Debashishs-MacBook-Pro utils_test % custom_grep -rn "TODO" .
1:TODO: implement more flags
```

3.4 custom_wc

Functionality: Performs word, line, and byte counts.

```
debashishbhagawati@Debashishs-MacBook-Pro utils_test % custom_wc fileA.txt fileB.txt
       5      20      97 fileA.txt
       1       5      30 fileB.txt
       6      25     127 total
debashishbhagawati@Debashishs-MacBook-Pro utils_test % custom_wc -L fileA.txt
      28 fileA.txt
```

3.5 custom_cp & custom_mv

Functionality: Handles file copying and moving. Implements verbose (`-v`) output to track progress.

```
debashishbhagawati@Debashishs-MacBook-Pro utils_test % custom_cp -vp fileA.txt fileA_backup.txt
'fileA.txt' -> 'fileA_backup.txt'
debashishbhagawati@Debashishs-MacBook-Pro utils_test % custom_cp -rv nested_dir backup_dir
'nested_dir/deep_dir/code.c' -> 'backup_dir/deep_dir/code.c'
debashishbhagawati@Debashishs-MacBook-Pro utils_test % custom_mv -v fileB.txt renamed_fileB.txt
'fileB.txt' -> 'renamed_fileB.txt'
debashishbhagawati@Debashishs-MacBook-Pro utils_test % custom_cp fileA.txt renamed_fileB.txt
debashishbhagawati@Debashishs-MacBook-Pro utils_test % custom_mv -iv fileA_backup.txt renamed_fileB
.txt
custom_mv: overwrite 'renamed_fileB.txt'?
y
```

3.6 custom_rm

Functionality: Deletes files and directories. Includes a recursive `-r` flag and an interactive `-i` flag for safety.

```
debashishbhagawati@Debashishs-MacBook-Pro utils_test % custom_rm -i renamed_fileB.txt
custom_rm: remove 'renamed_fileB.txt'? y
debashishbhagawati@Debashishs-MacBook-Pro utils_test % custom_rm -rfv backup_dir
removed 'backup_dir/deep_dir/code.c'
removed directory 'backup_dir/deep_dir'
removed directory 'backup_dir'
```

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4. Build Instructions

To compile the project, use the provided Makefile:

Compile all utilities

```
make all

# Install to /usr/local/bin

sudo make install

# Clean build files

make clean
```

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5. Conclusion

This project successfully replicates the core functionality of essential UNIX tools.